4 de Agosto de 2014

1) a) $X(z) = \log(1-2z)$, $|z| < 0{,}5$

b) $X(z) = \log(1-2z^{-1})$, $|z| > 2$

2) ROC $1-2z > 0 \Rightarrow 1 > 2z \Rightarrow z < \frac{1}{2}$

Entonces $\frac{\partial X(z)}{\partial z} = \frac{1}{1-2z} \cdot -2$

$n\,x[n] \xleftrightarrow{z} -z\frac{\partial X(z)}{\partial z}$ <= Prop. de la diferenciación.

$-z\frac{\partial X(z)}{\partial z} = \frac{z}{1-2z} \Rightarrow$

$1-2z = -2\left(-\frac{1}{2}+z\right)$

$-z\frac{\partial X(z)}{\partial z} = \frac{z}{-2\left(z-\frac{1}{2}\right)} = -\frac{1}{2}\cdot\frac{z}{z-\frac{1}{2}}$

↳ Antitransformo.

$n\,x[n] = -\frac{1}{2}\left(\frac{1}{2}\right)^n u[-n-1]$

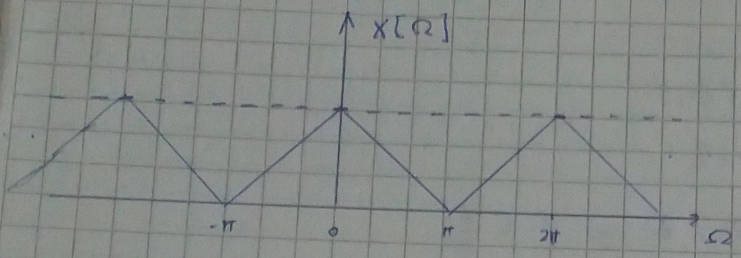
$x[n] = -\frac{1}{2n}\left(\frac{1}{2}\right)^n u[-n-1] = \frac{1}{n}\left(-\frac{1}{2}\right)^{n+1} u[-n-1]$

6) $\frac{\partial X(z)}{\partial z} = \frac{2z^{-2}}{1-2z^{-1}} \Rightarrow -z\frac{\partial X(z)}{\partial z} = \frac{-2z^{-1}}{1-2z^{-1}}$

$z^{-1}\left(\frac{-2}{z-2}\right) = -2\left(\frac{z^{-1}}{z-2}\right)$ <= $\boxed{z^{-1}} = (-2)(2)^{n-1}u[n-1]$

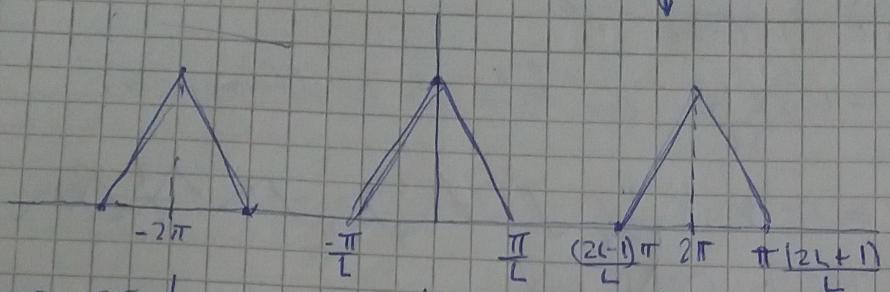
$x[n] = (-1)\frac{2^n}{n}u[n-1]$

2) Luego de pasar por el conversor:



Al realizar el upsampling

L↑



$2\pi - \frac{\pi}{L} = \frac{2\pi L - \pi}{L}$

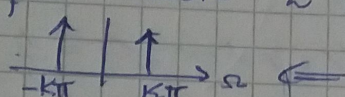
$\pi\frac{(2L-1)}{L}$

$2\pi + \frac{\pi}{L} = \frac{2\pi L + \pi}{L}$

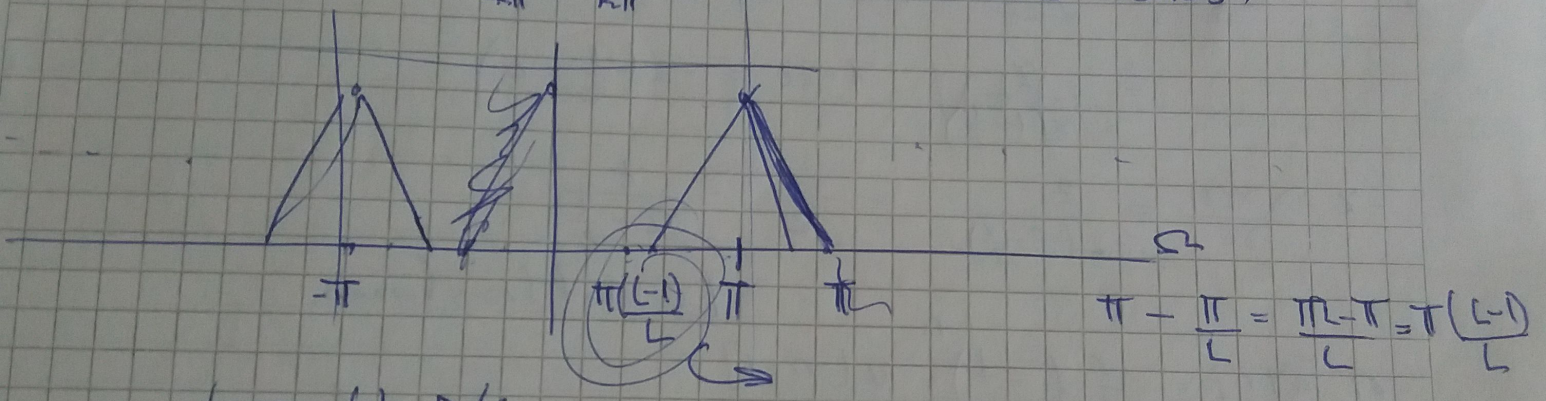
$\pi\frac{(2L+1)}{L}$

Necesito una señal que recorte 2 triángulos:

$\cos(\omega_0 t) \xrightarrow{F}$

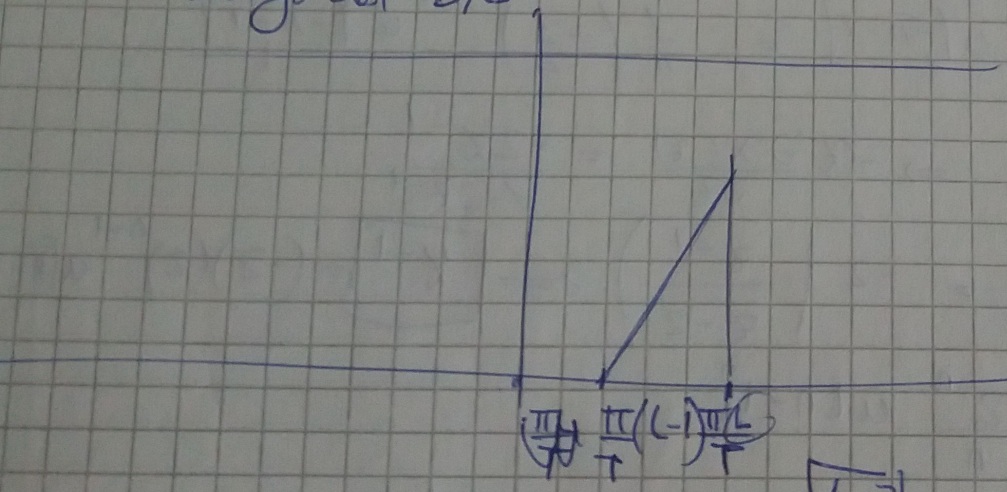


⇐ Convolucionar deltas:



$\pi - \frac{\pi}{L} = \frac{\pi L - \pi}{L} = \pi\frac{(L-1)}{L}$

Luego del D/C



$\Omega = \omega T$

$\pi = \omega \frac{T}{L}$

$\omega = \frac{\pi L}{T}$

$\pi\frac{(L-1)}{L} = \omega\frac{T}{L}$

$\frac{\pi}{T}(L-1)$

$\boxed{L=2}$